COMMERCEFLOW DATA ENGINEERING PLATFORM

FOCUSED RECONCILIATION AUDIT

Formal Production Audit, Medallion Lineage, Data Quality Verification, and Physical Performance Benchmarking

---

Target System	CommerceFlow Quick-Commerce Lakehouse
Primary Source Dataset	Brazilian E-Commerce Public Dataset by Olist (MIT License)
Audit Scope & Coverage	550,759 Raw Source Records • Date Span: 2016-09-04 to 2018-10-17
Reconciliation Status	100% Mathematical Parity • 0 Discrepant Records
Quality Check Score	10 of 10 Automated Quality Gates Passed (100.0%)
Physical Storage Engine	DuckDB 1.5.6 In-Process OLAP Lakehouse + SQLite 3 Transactional Metadata
Audit Generation Timestamp	2026-10-16 05:16:48 UTC
Security Classification	Strictly Traceable & Factual Engineering Evidence

Intended Audience & Purpose: *Designed dual-purpose for executive technical recruiters evaluating production data engineering rigor, and principal data architects inspecting exact SQL logic, schema grain, referential integrity, and microsecond query latencies. All figures are derived live without synthetic simulation.*

TABLE OF CONTENTS

Section	Title	Summary
Section 1	Executive Summary & Core Business Metrics	*High-level quick-commerce findings, dataset boundaries, and verified KPIs.*
Section 2	Dataset Provenance & Source Audit	*Canonical source URLs, licenses, SHA-256 fingerprints, and record counts.*
Section 3	Data Profiling & Quality Engine	*Column statistics, null counts, distinct values, and 10 quality rule evaluations.*
Section 4	Architecture & Medallion Data Flow	*Component topology, Bronze/Silver/Gold flow, and component rationale.*
Section 5	Data Lineage & Warehouse Modeling	*Star schema specification, grains, PK/FK relationships, and model lineage.*
Section 6	Transformation & SQL Engineering	*Deduplication rules, lead time calculations, null-handling, and SQL excerpts.*
Section 7	Business Analytics & Visual Findings	*Vector charts for category performance, monthly order volume, and logistics.*
Section 8	Reconciliation & Correctness Matrix	*Source vs warehouse queries, expected vs actual values, zero-variance proof.*
Section 9	Pipeline Operations & Reliability	*Run history, durations, controlled failure-injection, and safe recovery workflows.*
Section 10	Benchmarking & Physical Performance	*DuckDB analytical benchmark execution, P50/P95 latencies, and throughput.*
Section 11	Automated Testing & Verification	*24 backend pytest cases, 13 live E2E assertions, and verification criteria.*
Section 12	Security & Operational Boundaries	*Path traversal guards, read-only SQL sanitization, and upload defenses.*
Section 13	Technical Decisions & Trade-Offs (ADR)	*Why DuckDB, why medallion architecture, why event-time historical replay.*
Section 14	Honest Limitations & Future Work	*Known constraints, missing OCR, single-node boundaries, and future roadmap.*
Appendices	Data Dictionary, Glossary & Metadata	*Column glossaries, rule definitions, system glossary, and document hash.*

1. EXECUTIVE SUMMARY & PLATFORM OVERVIEW

1.1 Business Domain & Problem Statement

CommerceFlow addresses the fundamental data integrity challenge faced by modern e-commerce and quick-commerce operators: *Can we transform messy, unreliable, multi-source raw records into auditable, trustworthy business insights, and mathematically prove that the resulting data is correct?* Quick-commerce businesses operate under tight logistics margins where discrepancies between inventory manifests, fiscal invoices, order heads, and payment settlements directly erode profitability.

1.2 Dataset Distinctions & Boundaries

The platform is engineered around the authentic **Brazilian E-Commerce Public Dataset by Olist**. It encompasses **99,441 verified orders** spanning from September 2016 through October 2018 across 27 Brazilian federative units. Crucial boundary distinction: This report and platform explicitly document historical marketplace logistics; historical records are never misrepresented as live GPS vehicle telemetry. Historical event replay is decoupled into **Event Time** (original purchase timestamp) and **Ingestion Time** (current system timestamp) to study out-of-order event arrivals.

1.3 Key Verified Platform Metrics

Total Ingested Orders	99,441	**Total Line Items Sold**	112,650
Gross Merchandising Value	R$ 15,848,243.14	**Delivered Order Count**	96,478 (97.0%)
Delivered Settlement Sales	R$ 15,422,461.77	**Quality Gate Pass Rate**	100.0% (10 of 10 Passed)
Data Reconciliation Parity	0 Variance (100% Match)	**DuckDB OLAP P50 Latency**	60.90 ms

8. RECONCILIATION & MATHEMATICAL CORRECTNESS

A core tenet of CommerceFlow is auditability. Below is the live reconciliation matrix proving mathematical parity between raw CSV inputs and warehouse dimensional outputs:

Reconciled Metric	Source Value	Warehouse Value	Variance	Parity Status	Auditor Explanation
Total Order Record Count	99,441	99,441	0	**100% MATCH**	*Perfect 1:1 row preservation through Medallion bronze, silve...*
Total Order Line Items	112,650	112,650	0	**100% MATCH**	*Every valid line item preserved; duplicate keys quarantined ...*
Customer Header Dimension	99,441	99,441	0	**100% MATCH**	*Zero customer records dropped; ZIP prefix geolocation mapped...*
Catalog Product Dimension	32,951	32,951	0	**100% MATCH**	*Enriched with English category translation dictionary....*
Gross Item Merchandising Value	R$ 13,591,643.70	R$ 13,591,643.70	0.0	**100% MATCH**	*Sum of line item prices matches aggregated header total_item...*

Zero-Drop Guarantee: *All variance counts evaluate to 0, certifying that transformations neither duplicated nor silently discarded valid transactions.*

APPENDIX: DATA DICTIONARY & GLOSSARY

Metric / Field	Entity	Definition & Computational Semantics
`delivery_days`	fact_orders	Calendar days elapsed between order_purchase_timestamp and order_delivered_customer_date.
`estimated_delivery_margin_days`	fact_orders	Difference between promised delivery date and actual customer delivery date. Positive margin indicates early delivery.
`total_order_cost`	fact_orders	Total financial value of the order computed as sum of item prices plus freight charges.
`reconciliation_variance`	fact_orders	Absolute difference between total order cost and total recorded customer payment values.

Report Generation Stamp: *Document compiled at 2026-10-16 05:16:48 UTC using ReportLab 4.4.10 under Python 3.14.7. All values derived from persisted warehouse lakehouse states.*